

LLM-Based Crypto Market Signal Detection from X

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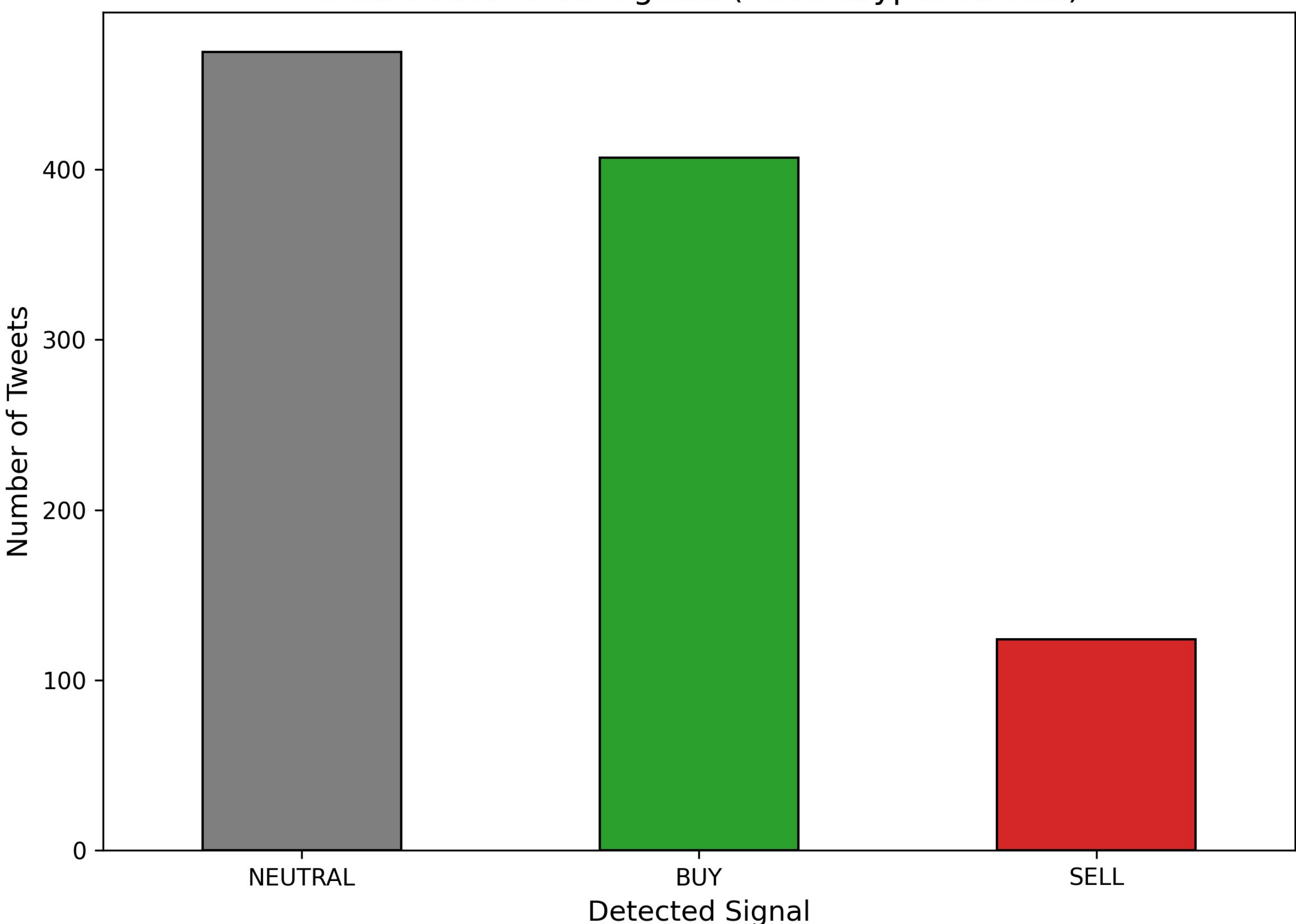
Motivation

- **Market Vulnerability:** While cryptocurrencies are popular investments, the markets are highly unpredictable and erratic.
- **Social Media Impact:** Public figures and influencers possess considerable power over market trends, where a single social media post can trigger financial shifts
- **Project Objective:** The goal is to apply Large Language Models (LLMs) to filter influential posts and evaluate their exact impact on the market.

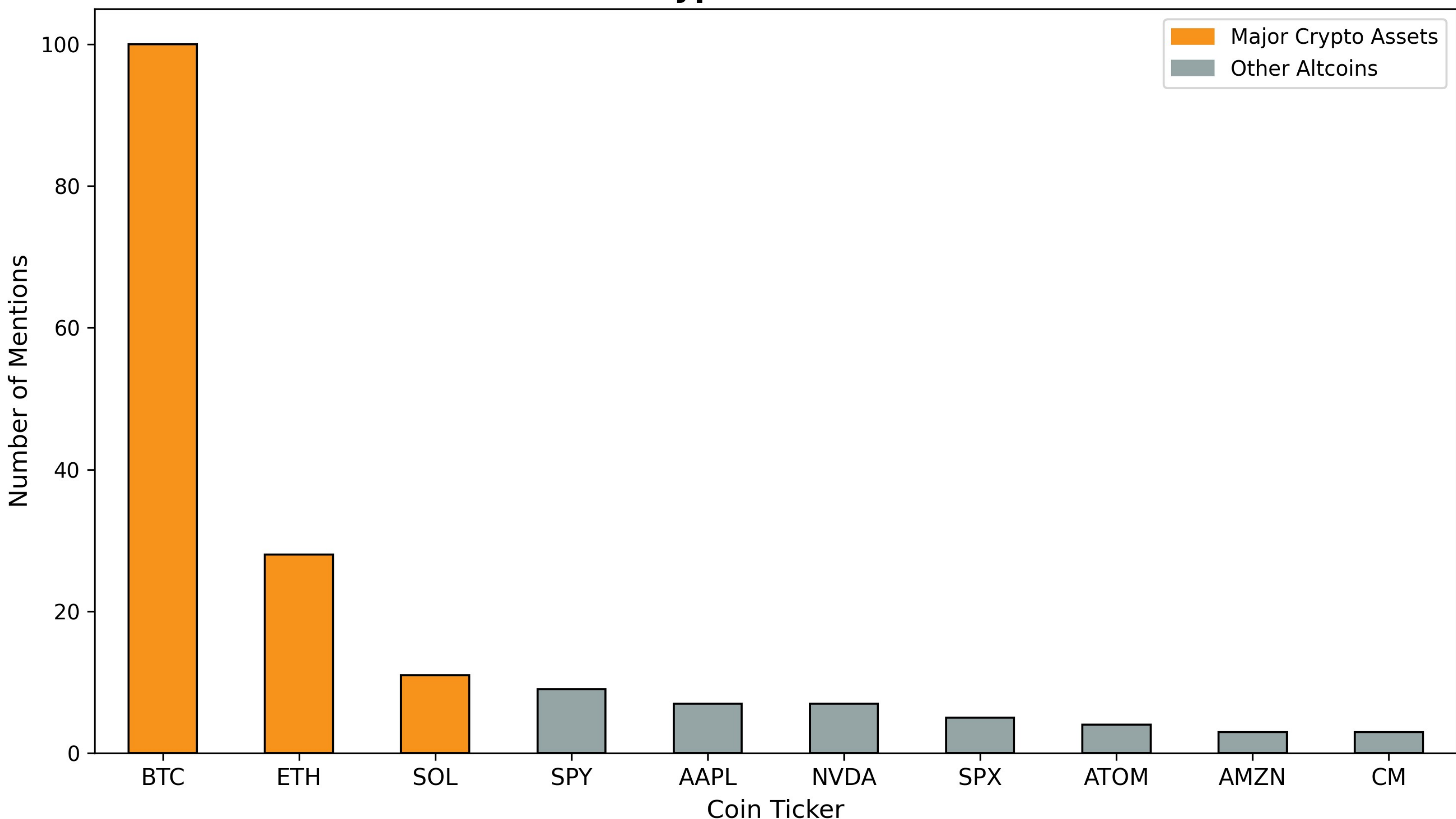
Implementation

- **Data Source:** Tweets are sourced from public datasets and analyzed by Llama 3.1 for sentiment
- **AI Analysis (Llama 3.1):** We send the text of each tweet to an AI model that reads it
- **Result:** The AI classifies the sentiment: Is it positive ("BUY") , neutral ("NEUTRAL") or negative ("SELL")?
- **Detection:** Identifying posts that could potentially trigger market movements (e.g. Like/Retweet counts)
- **Confirmation:** Evaluating historical price data to verify whether a specific post actually caused a measurable impact on the cryptocurrency market.

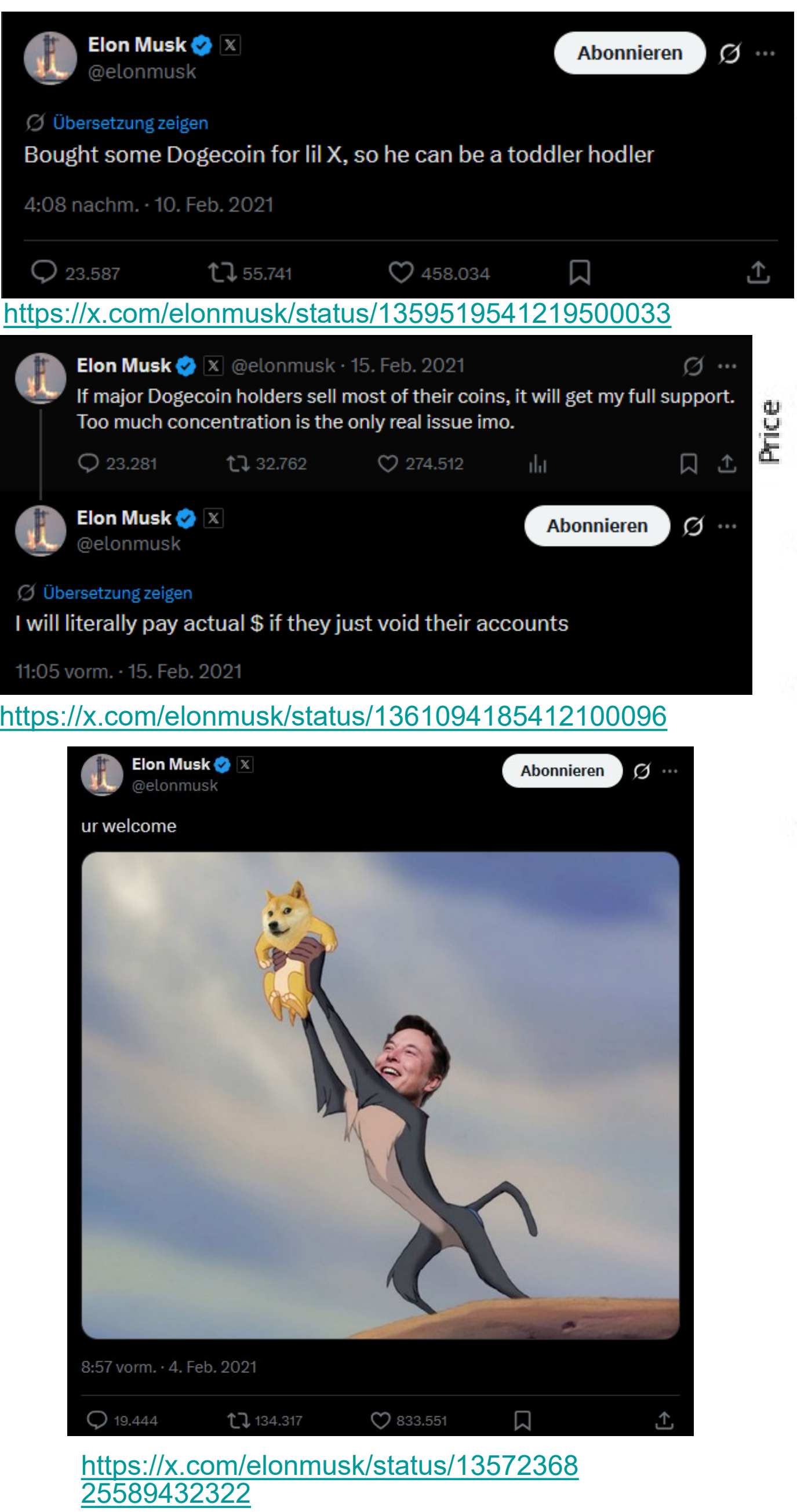
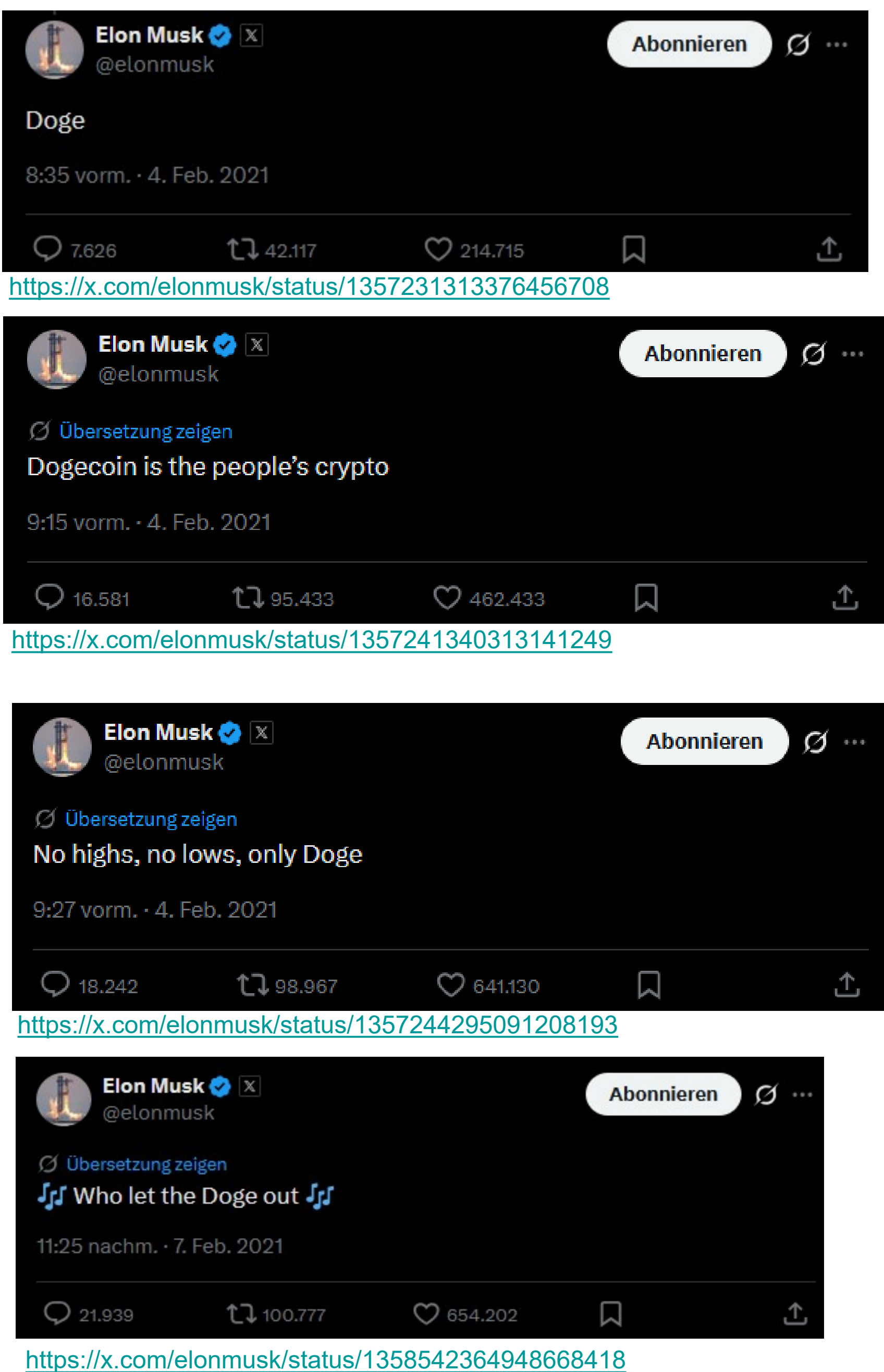
Distribution of AI Signals (1000 Crypto Tweets)



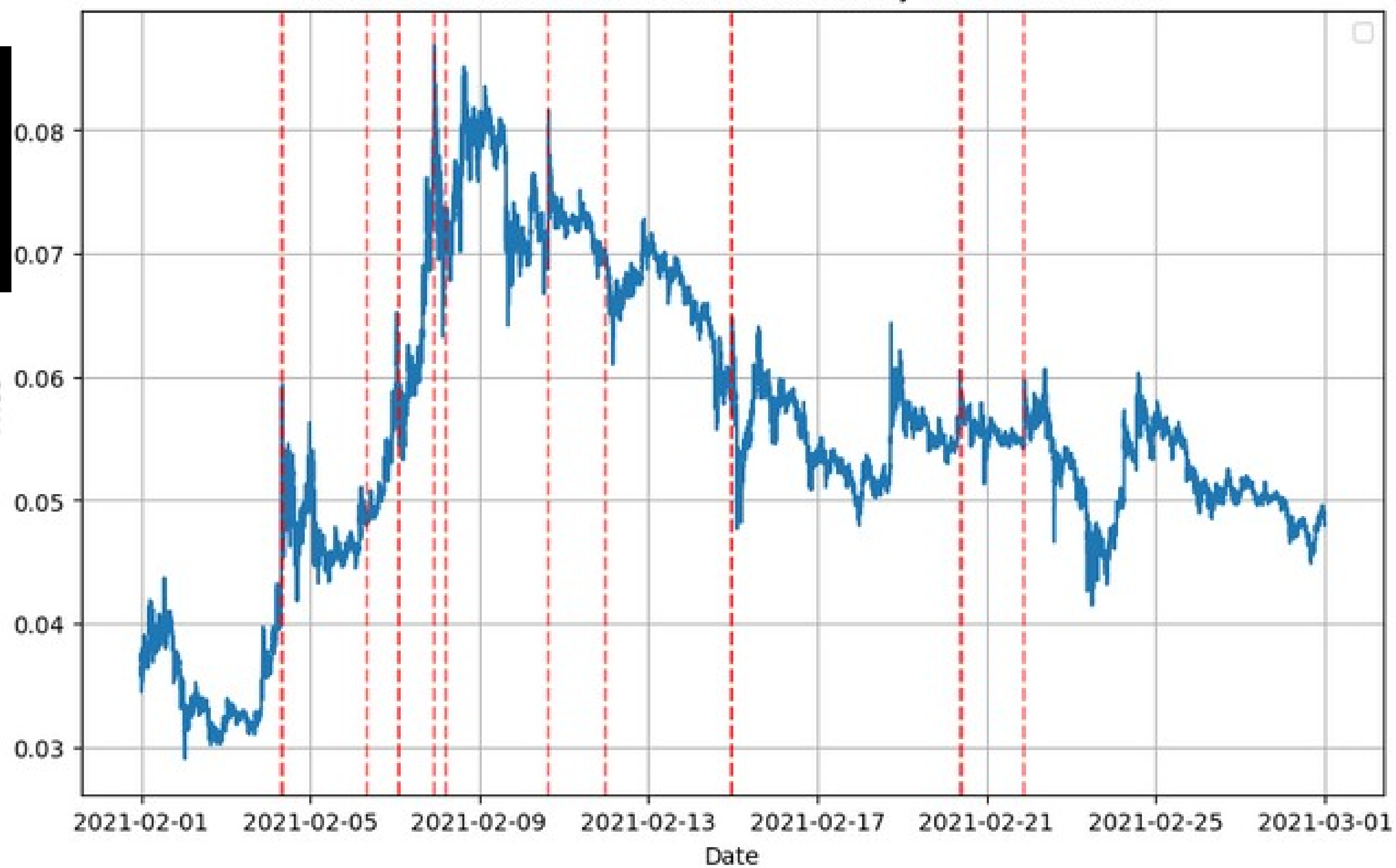
Most Mentioned Cryptocurrencies in 1000 Tweets



Elon Musk's Influence on Dogecoin



Time Series Plot of DOGE Prices in February with Tweet Dates



- **Correlation:** Significant price changes often happened right after popular tweets

Resources

[ABC20] Stephan Akkerman. "Financial Tweets Dataset." Hugging Face, 2023. <https://huggingface.co/datasets/StephanAkkerman/financial-tweets>
[ABC21] Mendeley Data. "Cryptocurrency Sentiment Analysis Dataset." Mendeley Data, 2020. <https://data.mendeley.com/datasets/x7yvshrmxy/1>
[ABC22] Jean (johnsmith44). "Musk Doge tweets and effect on price, Feb 2021" <https://www.kaggle.com/code/johnsmith44/musk-doge-tweets-and-effect-on-price-feb-2021>
[ABC23] Elon Musk. "Twitter/X Tweets from Feb 2021" <https://x.com/elonmusk>